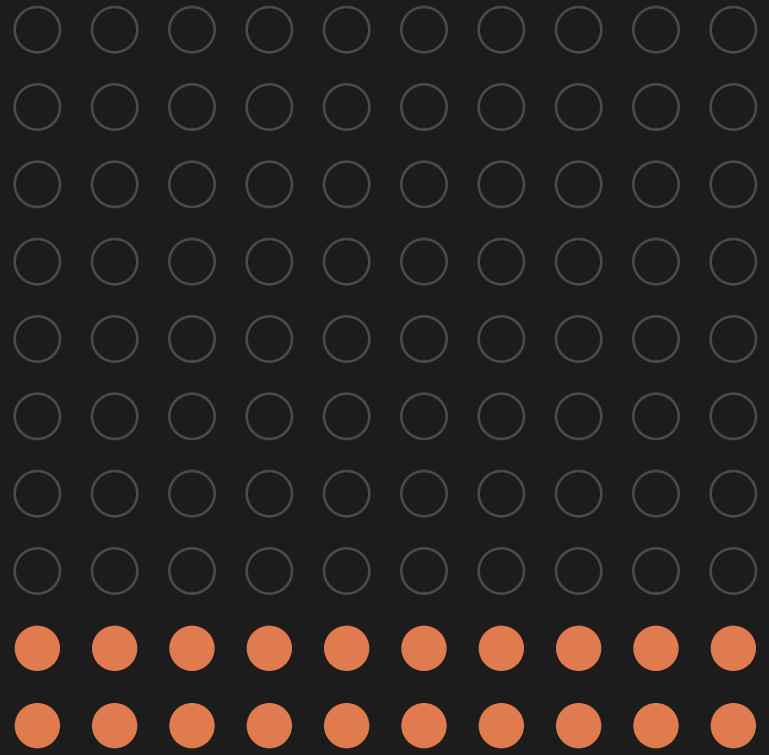
NO-SHOW ANALYSIS

How to reduce no-shows to medical appointments

Diagnosis of the factors associated with no-shows and recommendation of an SMS pilot with active confirmation



20.2%

of appointments ended in a no-show: 1 in 5

Executive presentation

2016 historical data · 110,521 appointments

NO-SHOW ANALYSIS

Lead time drives no-shows; reminders can help

PROBLEM

Medical appointments

20.2%

The historical dataset showed a no-show rate of **20.2%**, representing **22,314 no-shows out of 110,521 valid appointments**.

KEY FINDINGS

4.6% → 33.0%
same day → 31+ days

Lead time: the no-show rate ranged from 4.6% for same-day appointments to 33.0% for appointments booked 31 or more days ahead.

SMS
4+ day groups

SMS: within comparable lead-time groups, appointments with an SMS had lower no-show rates in groups of four days or more.

26.9% vs 18.8%
with vs without prior no-show

No-show history: patients with a recorded prior no-show had a rate of 26.9%, against 18.8% among returning patients with no prior no-show.

RECOMMENDATION

Test an **SMS process with active confirmation**, allowing patients to confirm or cancel in advance, and assess how freed-up slots are reused.

NEXT STEP

Propose a **controlled pilot** to measure the effect on the no-show rate, slot utilization and the costs involved.

Note: the data are from 2016. The associations found do not prove causation, and the impact of the solution still needs to be tested.

How can we reduce no-shows to medical appointments?

01



The health unit **schedules appointments** to meet patient demand and use its care capacity. **Attendance** is what makes those schedules productive.

02



Some patients **miss their scheduled appointments**, leaving capacity idle. It is still **unclear which factors** are most associated with no-shows, or which actions could reduce them.

03



Which factors are associated with no-shows, and ***which initiatives should be prioritized*** *to reduce no-shows and make better use of care capacity?*

20.2%

Size of the problem in the dataset

of recorded appointments ended in a no-show (2016 historical data)

Note: the case brief cites a 30% no-show rate; the dataset shows 20.2%, adopted as the baseline. Source: Medical Appointment No-Shows (Kaggle), 2016.

PROBLEM AND OBJECTIVE

One in five appointments is missed, leaving slots empty

110,521
valid appointments in the historical data

22,314
ended in a no-show

20.2%
Baseline: no-show rate



Breakdown of the 110,521 valid appointments

OBJECTIVE

Identify the **factors associated with no-shows** and **prioritize initiatives** that can reduce no-shows and make better use of care capacity.

APPROACH

1. Structure **hypotheses**
2. Test them with the **available data**
3. Turn the main findings into **practical recommendations**

SCOPE AND LIMITATION

The analysis uses **2016 data** and identifies **associations**; the effectiveness of the proposed solutions **still needs to be validated**.

Source: Medical Appointment No-Shows (Kaggle), 2016.

HYPOTHESIS TREE

Four factors were analyzed to identify what drives no-shows

CENTRAL QUESTION *What sets apart the appointments patients attend from the ones they miss?*

1 Scheduling

Lead time
Is a longer lead time between booking and appointment associated with more no-shows?

Weekday
Does the weekday affect the no-show rate?

Lead-time groups
Are no-shows concentrated in specific lead-time groups?

2 Communication

SMS
Do appointments with an SMS have a lower no-show rate?

SMS × lead time
Does the association between SMS and no-shows hold when controlling for lead time?

3 Patient profile

Age and gender
Are age and gender associated with no-shows?

Conditions and indicators
Are Bolsa Família, hypertension, diabetes, alcoholism and disability associated with no-shows?

No-show history
Is no-show history associated with future attendance?

4 Location

Neighborhood
Are there neighborhoods with above-average no-show rates?

Neighborhood × lead time
Do the differences between neighborhoods hold after controlling for lead time?

EVALUATION CRITERION

Size of the difference, statistical evidence and operational relevance.

EXPECTED OUTCOME

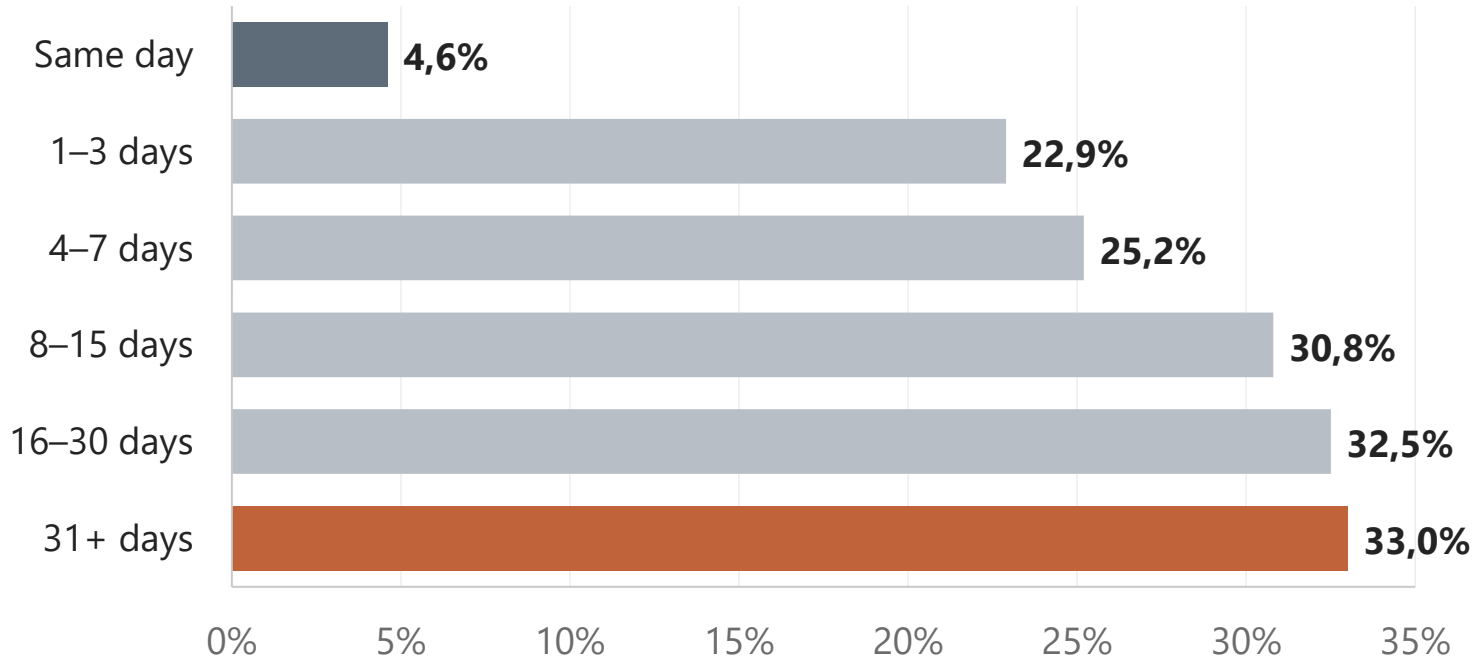
Identify priority factors, rule out hypotheses with no practical relevance and define initiatives worth testing.

ANALYSIS · SCHEDULING

Booking lead time is associated with higher no-show rates

Appointments booked further ahead have higher no-show rates.

No-show rate by booking lead time, %



KEY INSIGHTS

28.4 p.p.
difference between the no-show rate of same-day appointments and those booked 31 or more days ahead.

92% of no-shows
occur in appointments booked at least one day ahead.

Cramér's V = 0.295
moderate association between lead time and no-shows.

CONCLUSION *Finding ways to reduce no-shows in appointments booked in advance should be a priority.*

Note: historical data from April to June 2016. The results show association, not causation. Same-day appointments may have different operational characteristics.

ANALYSIS · COMMUNICATION

SMS is associated with lower no-show rates when we control for lead time

The direct comparison is biased by differences in the profile of appointments that receive an SMS

1 RAW COMPARISON

16.7%
Without SMS

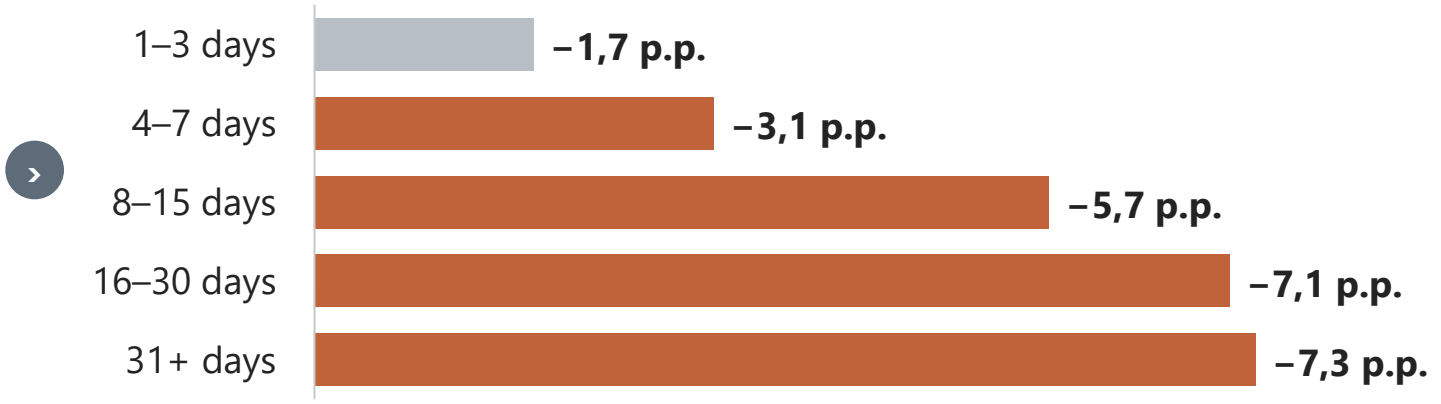
27.6%
With SMS

Raw difference: **+10.9 percentage points**

At first sight, appointments with an SMS seem to have more no-shows. However, messages are sent mainly to appointments with **longer lead times**, which already have higher no-show rates.

2 COMPARISON WITHIN THE SAME LEAD-TIME GROUP

Difference in no-show rate associated with SMS, by lead time, p.p.



Negative = fewer no-shows with SMS. At 1–3 days, the difference is not significant (p = 0.24).

MAIN CONCLUSION

With similar lead times, receiving an SMS is associated with **lower no-show rates in groups of 4 days or more**.

BUSINESS IMPLICATION

Prioritize an **SMS pilot with active confirmation** for appointments booked further ahead, assessing its effect against a control group.

Methodological note: an observational association does not prove causation. The result may be influenced by other differences between the groups.

ANALYSIS · PROFILE AND LOCATION

No-show history and location help identify priority groups

Rates vary by history and neighborhood; for neighborhoods, even after controlling for lead time

1 NO-SHOW HISTORY

18.8%
No recorded prior no-show

26.9%
Recorded prior no-show

Difference: **+8.1 p.p.**

Patients with a **recorded prior no-show** have a higher no-show rate in the dataset.

2 DIFFERENCES BETWEEN NEIGHBORHOODS

Six neighborhoods had rates at least **3 p.p. above the overall average of 20.2%**:



The differences remained positive after controlling for lead time.

3 OTHER FACTORS

Age

26.0% | 23.1%
highest among teenagers and young adults

Bolsa Família

+3.9 p.p.
compared with non-beneficiaries

Gender

0.3 p.p.
small difference

BUSINESS IMPLICATIONS

Test **additional reminders** for patients with prior no-shows, assessing the result against a control group.

Investigate possible **access and communication barriers** in high-rate areas, without penalizing residents.

Use the results as **signals for investigation**, not as proof of individual causes.

Methodological note: the available history covers only three months. The observed differences do not prove causation and may reflect factors not recorded in the data.

Not every factor is relevant enough to guide action

The analysis prioritizes factors with meaningful differences and operational potential

PRIORITIZE FOR INVESTIGATION

Confirmed hypotheses or relevant associations

- **Booking lead time:** longer lead time associated with a higher no-show rate.
- **SMS controlled for lead time:** associated with lower no-show rates in appointments booked 4 or more days ahead.
- **No-show history:** 26.9% with a recorded prior no-show, against 18.8% among returning patients without one.
- **Age:** teenagers and young adults have higher rates.
- **Neighborhoods:** six neighborhoods above the defined criterion, even after controlling for lead time.
- **Scholarship and hypertension:** smaller associations; interpret with caution.

NO SPECIFIC ACTION FOR NOW

Not practically relevant enough, or inconclusive

- **Weekday and gender:** small differences.
- **Diabetes, alcoholism and disability:** did not meet the practical relevance criterion set in the analysis.
- **Concentration in one or two groups:** hypothesis rejected under the adopted criterion.
- **SMS without controlling for lead time:** the raw comparison is potentially biased; it should not be used alone to assess the effect of the message.

DECISION *Test a **communication intervention** before scaling up more complex actions.*

PRIORITIZATION

Prioritize communication, then test other actions

Solutions should act at different stages of the booking journey

1 BEFORE THE APPOINTMENT
Reduce no-shows

PRIORITY 1

SMS with active confirmation

- Send a message asking the patient to confirm or cancel.
- Initial pilot with appointments booked 4 or more days ahead.
- Measure the effect against a control group.

2 WHEN THE PATIENT CANNOT ATTEND
Recover the slot

PRIORITY 2

Waiting list

- Contact available patients when there is a cancellation.
- Prioritize by clinical and operational criteria.
- Measure how many freed-up slots are filled.

3 PATIENTS WITH A NO-SHOW HISTORY
Test additional contact

PRIORITY 3

Targeted reminder

- Additional contact for patients with a recorded prior no-show.
- Test with a control group.
- Measure the reduction in no-shows and the cost per no-show avoided.

4 INVESTIGATION TRACK
Differences between neighborhoods

Assess **SMS coverage, access, transport and communication** and validate the possible explanations before proposing specific interventions.

RECOMMENDATION Start with the **SMS pilot with active confirmation**. The other initiatives should be assessed in parallel or in later stages, depending on the operational data available.

PRIORITIZATION CRITERION
Evidence, impact, operational effort, cost and feasibility.

RECOMMENDATION

Test SMS with active confirmation

WHY PRIORITIZE IT?

3.1 to 7.3 p.p.

fewer no-shows with SMS in groups of 4 days or more

Appointments with **longer lead times** have higher no-show rates.

Within groups of 4 days or more, appointments **with an SMS** had no-show rates 3.1 to 7.3 percentage points lower.

The observed association **justifies a test**, but does not yet prove a causal effect.

HOW TO VALIDATE?

1. **Controlled pilot**
   Run a pilot with an intervention group and a comparable control group.
2. **Rules set in advance**
   Define the duration, eligible population and success target beforehand.
3. **Comparison between groups**
   Compare no-show rates between the groups.
4. **Full evaluation**
   Assess costs, responses, early cancellations and reused slots.

EXPECTED DECISION

Scale up, adjust or stop the initiative based on the results.

Scale up | **Adjust** | **Stop**

The pilot is a proposal. The real effect, cost and reduction target still need to be validated.

NEXT STEPS

Validate the impact before scaling up the initiatives

Three next steps to turn findings into results

01 VALIDATE

Validate the current scenario

Update the analysis with **recent data** and confirm whether the no-show patterns persist.

02 TEST

Test the main recommendation

Run an **SMS pilot with active confirmation**, comparing the results with a control group.

03 DECIDE

Decide based on evidence

Assess the **reduction in no-shows**, the **cost per no-show avoided** and operational feasibility before scaling up the initiative.

EXPECTED DECISION

*Move forward with the pilot, measure the results and **expand only if there is a proven benefit**.*

The data analyzed are from 2016; the current impact needs to be validated.

Turning data into action to reduce no-shows

Lead time is the main risk signal identified; communication is an opportunity to validate

20.2%

Historical no-show rate: 22,314 no-shows out of 110,521 appointments.

33.0%

No-show rate for appointments booked 31 or more days ahead.

−3.1 to −7.3 p.p.

Differences in no-show rates associated with SMS in groups booked 4 or more days ahead.

EXECUTIVE RECOMMENDATION

Validate current data and **test SMS with active confirmation**, comparing the results with a control group before scaling up the initiative.

NEXT STEP

Run a **controlled pilot** and assess the reduction in no-shows, cost per no-show avoided and operational feasibility.

There is an opportunity to improve attendance; scaling up the initiative depends on a test with current data.

Historical data from April to June 2016. The results are observational and do not prove causation.

APPENDIX

Appendix

Supporting detail: methodology, calculations and reference tables

CONTENTS

A1 Data and cleaning

A2 Statistical approach

A3 Results by hypothesis

A4 SMS by lead-time group

A5 Neighborhoods × lead time

A6 Risk segmentation

A7 Limitations

A8 Resources

Historical data from April to June 2016. The results are observational and do not prove causation.

APPENDIX · A1 · DATA AND CLEANING

Data and cleaning

110,521
valid appointments (final base)

22,314
ended in a no-show

20.2%
baseline no-show rate

SOURCE & PERIOD

- **Kaggle —** "Medical Appointment No Shows", a public dataset of public-health bookings
- **Location:** Vitória, Espírito Santo, Brazil
- **Period:** 29 Apr – 8 Jun 2016 (about six weeks of history)

RECORDS REMOVED

- **Age = −1** — invalid entry
- **Negative lead time** — appointment dated before the booking
- **Final analysis base:** 110,521 valid appointments

Source: Kaggle — Medical Appointment No Shows (Vitória / ES, Brazil, 2016). Figures reflect the cleaned analysis base.

Statistical approach

THE TEST

Chi-square (χ^2) test of independence.

Compares the no-show rate across the groups of a variable (e.g. lead-time bands) and asks whether the differences are larger than chance would produce.

DECISION RULE

A hypothesis is kept only if both hold:

$p < 0.05$
statistically significant

difference ≥ 3 p.p.
practically relevant

The size threshold stops a large sample from flagging trivial gaps.

EFFECT SIZE — CRAMÉR'S V

Measures how strong an association is, on a 0–1 scale.

- **≈ 0.1** weak
- **≈ 0.3** moderate
- **≈ 0.5 +** strong

Reported next to every significant test (e.g. lead time, V = 0.295).

Associations are reported, not causal effects. Thresholds ($p < 0.05$; ≥ 3 p.p.) were fixed before testing.

Results by hypothesis

Hypothesis	Key difference	Verdict
H1.1 Lead time	Same-day 4.6% → 31+ days 33.0% (Δ 28.4 p.p.; V = 0.295)	**Prioritize**
H1.2 Weekday	Small difference (< 3 p.p.)	No action
H1.3 Lead-time groups	No-shows not concentrated in one or two groups	No action
H2.1 SMS (raw)	Raw +10.9 p.p. — biased by appointment profile*	No action
H2.2 SMS × lead time	−1.7 to −7.3 p.p. fewer no-shows (4+ days ahead)	**Prioritize**
H3.1 Age & gender	Age: teens & young adults higher; gender 0.3 p.p.	**Prioritize**
H3.2 Conditions & indicators	Bolsa Família +3.9 p.p.; others below 3 p.p.	**Caution**
H3.3 No-show history	18.8% → 26.9% (+8.1 p.p.)	**Prioritize**
H4.1 Neighborhood	Several neighborhoods above the average rate	**Prioritize**
H4.2 Neighborhood × lead time	6 neighborhoods above criterion after controlling	**Prioritize**

Both criteria apply: $p < 0.05$ and difference ≥ 3 p.p. *Raw SMS comparison is biased by lead-time profile (see A4). Full per-hypothesis Cramér's V in the README (A8).

SMS by lead-time group

No-show rate with SMS minus without SMS, within each group (negative = fewer no-shows)

Lead-time group	SMS effect	Significance
31+ days	**−7.3 p.p.**	Significant
16–30 days	**−7.1 p.p.**	Significant
8–15 days	**−5.7 p.p.**	Significant
4–7 days	**−3.1 p.p.**	Significant
1–3 days	**−1.7 p.p.**	Not sig. (p = 0.24)

WHY THE AGGREGATE REVERSES

+10.9 p.p.

raw gap, with SMS (27.6%) vs without (16.7%) — SMS looks worse

SMS is sent far more often for appointments booked well ahead — the highest-risk group. Pooling the groups makes SMS look harmful, **even though it is linked to fewer no-shows within every comparable group.**

This reversal is Simpson's paradox.

Chi-square per group. Negative = fewer no-shows with SMS. The 1–3 day group is not statistically significant (p = 0.24).

Neighborhoods × lead time

Hypothesis H4.2 — do neighborhood differences hold after controlling for lead time?

6

neighborhoods stay above the no-show criterion even after controlling for lead time

HOW IT WAS TESTED

Appointments are split into the five lead-time bands. Within each band, every neighborhood's no-show rate is compared against the same practical-relevance criterion — so longer booking lead times in an area cannot, on their own, explain a high rate.

WHAT IT MEANS

Six neighborhoods stay above the criterion inside comparable bands. Their higher no-show rates are **not just a by-product of lead time** — location is an independent signal worth targeting.

Per-neighborhood no-show rates by lead-time band are available in the Power BI dashboard (A8).

Risk segmentation

The confirmed drivers — prior no-show history, long lead time, younger age — combine into three risk bands

HIGH RISK

A prior no-show on record and a long booking lead time; skews toward younger patients.

MEDIUM RISK

One strong signal — either a prior no-show **or** a long lead time, but not both.

LOWER RISK

Same-day or short lead time and no prior no-show; attendance is already high.

41%
of all appointments

63%
of all no-shows

sit in the high- and medium-risk bands — the groups to target first with reminders and active confirmation.

Bands combine the confirmed drivers; exact thresholds and per-band figures are defined in the README and dashboard (A8).

Limitations

- **Observational data.** associations, not causation. We cannot say an SMS or reminder causes attendance — unmeasured factors may drive both.
- **Short window.** about six weeks of history (Apr–Jun 2016). Seasonality and year-over-year trends are not captured.
- **No reason recorded.** illness, transport, work or simple forgetfulness are indistinguishable in the data.
- **Single city and year.** Vitória / ES, 2016. Findings may not transfer to other services or periods.
- **Partial history.** a prior no-show is only observable for returning patients; first-time patients have no history.

Limitations are scoping notes for interpreting the findings, not reasons to discount them.

Resources

CODE & DOCUMENTATION

GitHub repository

Full analysis and data-cleaning steps, with the README documenting **every hypothesis test** — difference and Cramér's V.

REPOSITORY

Open on GitHub ↗

CALCULATIONS

Excel workbook

The statistics behind the deck: **chi-square tests, differences and Cramér's V** by hypothesis, group and neighborhood.

FILE

Open in Google Sheets ↗

INTERACTIVE DASHBOARD

Power BI

No-show rates by lead time, SMS, patient profile, neighborhood and risk band — **filterable for exploration.**

DASHBOARD

Open in Power BI ↗

Source: project analysis of the Kaggle Medical Appointment No Shows dataset (Vitória / ES, Brazil, 2016).